APPM 4600: HW8 P.2

2. Interpolation Adventures

Interpolate $F(x) = \frac{1}{1+x^2}$ on $[-5,5]$. Using equispaced nodes w/ $n = 5, 10, 15,$ and 20, interpolate using:

a. Lagrange
b. Hermite
c. Natural Cubic
d. Clamped Cubic.

Which performs best? why?

~~a. Lagrange:~~

a. For $n=20$ Lagrange and Hermite on equispaced nodes suffer from increasing error near either side of our interval. Splines perform better in terms of absolute error by as much as 6 orders of magnitude in the extremes. Near the centre they provide comparable performance.

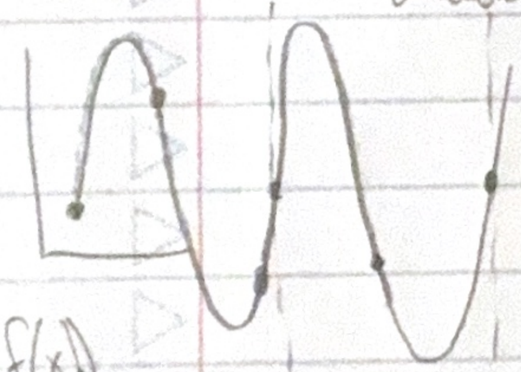
b. Chebyshev nodes drastically increase performance of Lagrange / Hermite near the sides, but appear to have little impact on spline, probably b/c splines are defined locally so don't see end effects.

For $n=5$, hermite/Lagrange and both natural and clamped splines suffer equal error in the $[0,2]$ range, except at specific points where they match $f(x)$ exactly. Chebyshev reduces all error near the sides, but not near centre.

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4. Approx. a function like $f(x) = \sin(10x)$ on $[0, 2\pi]$.

To make these splines periodic, I propose that the end point conditions at the conclusion of each period match those of the next.



This would look like:

$$S_{n+1}(x_n) = S_{n+1}(x_n + P)$$
$$S'_{n+1}(x_n) = S'_{n+1}(x_n + P) = f'(x_n)$$
$$S''_{n+1}(x_n) = S''_{n+1}(x_n + P) = M_{n+1} \neq 0$$

As a project I will code this.